

Hallucinations in LLMs: Types, Causes, and Approaches for Enhanced Reliability

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Abstract—Large Language Models (LLMs) have transformed natural language processing and generation, enabling diverse applications ranging from content creation to decision support systems. However, a key challenge they face is hallucination—the generation of content that is factually incorrect, inconsistent, or entirely fabricated, yet often appears coherent and plausible. This survey comprehensively examines hallucinations in LLMs, categorizing them into intrinsic, extrinsic, amalgamated, and non-factual types, and analyzing underlying mechanisms such as knowledge overshadowing and contextual misalignment. We evaluate the impact of hallucinations in critical domains like healthcare, scientific research, and journalism, where misinformation can have serious consequences. On the other hand, we explore the potential for hallucinations to drive creativity in fields like art and design, where unconventional outputs can inspire innovation. The survey also reviews detection methods, including named entity recognition and probability-based approaches, alongside mitigation strategies such as prompt engineering, fine-tuning, and grounding techniques. Additionally, we discuss future research directions, focusing on improved evaluation methods, ethical considerations, and advanced integration techniques to enhance the reliability and ethical use of LLMs. By balancing accuracy with creativity, this survey aims to contribute to the development of more trustworthy and responsible large language models.

I. INTRODUCTION

Large Language Models (LLMs) have brought significant advancements to artificial intelligence, particularly in natural language processing and generation. However, one of the most prominent challenges they face is hallucination, where the model produces content that is factually incorrect, inconsistent, or entirely fabricated, despite appearing coherent and plausible [1]–[5].

Hallucinations in LLMs are of increasing concern as these models become integral to applications such as content creation and decision support systems, where the accuracy of generated information is crucial [6]. These hallucinations can lead to misinformation, incorrect conclusions, or flawed decision-making in critical areas such as healthcare, scientific research, and journalism [7]–[10]. Interestingly, some recent studies suggest that hallucinations may not be wholly detrimental. Their potential to inspire novel ideas and generate unexpected connections could be beneficial in certain creative contexts, such as brainstorming or ideation, even if these outputs are not grounded in fact [1], [11].

Addressing hallucinations is now a priority in AI research, with ongoing efforts to detect, reduce, and even harness these phenomena while maintaining high standards of factual accuracy [7]. The challenge lies in improving LLMs to retain their language generation capabilities while minimizing harmful hallucinations and exploring possible productive uses of creative outputs [11].

This survey reviews current research on hallucinations in LLMs, including their types, causes, impacts, and mitigation strategies. By examining both the challenges and potential advantages, we aim to contribute to the discussion on enhancing the reliability and utility of LLMs, with a focus on balancing accuracy and creativity.

II. TYPES AND CAUSES OF HALLUCINATIONS

Hallucinations in Large Language Models (LLMs) are a growing concern in artificial intelligence, where these models generate outputs that deviate from factual information or the given context. This section outlines the taxonomy of hallucinations in LLMs and examines the underlying mechanisms contributing to this phenomenon.

A. Taxonomy of Hallucinations

Hallucinations in LLMs can be categorized into several distinct types:

1) *Intrinsic Hallucinations*: Intrinsic hallucinations result from limitations in the model’s internal knowledge. These occur when the LLM generates outputs inconsistent with the information encoded in its training data or general world knowledge. This type of hallucination typically arises from the model’s failure to accurately retrieve or represent information from its parametric memory [12].

2) *Extrinsic Hallucinations*: Extrinsic hallucinations occur due to issues with integrating external information. These arise when the model misinterprets or fails to incorporate the given context or prompt correctly. In multimodal models, this may manifest as visual hallucinations, where the model fabricates details about an image in tasks like visual question answering [13].

3) *Amalgamated Hallucinations*: Amalgamated hallucinations emerge when the model incorrectly combines multiple facts or conditions. These occur when a prompt presents several conditions, and the model fails to integrate them properly, resulting in a blended output that erroneously merges disparate pieces of information [14].

4) *Non-Factual Hallucinations*: Non-factual hallucinations involve the generation of content that contradicts established facts or world knowledge. This type of hallucination is especially problematic in knowledge-intensive tasks, where it can propagate misinformation [15].

B. Underlying Mechanisms of Hallucinations

Several underlying mechanisms contribute to hallucinations in LLMs:

1) *Knowledge Overshadowing*: Knowledge overshadowing occurs when certain aspects of a prompt dominate the model's attention, leading to hallucinations. This phenomenon is partly due to imbalances in the training data and can be understood as an overgeneralization of dominant conditions or patterns [14].

2) *Insufficient Knowledge Representation*: Hallucinations can arise from insufficient knowledge representation, particularly in the lower layers of the model's neural network. These deficiencies result in what is termed "knowledge enrichment hallucinations," where the model generates unsupported information due to gaps in its subject-specific knowledge [15].

3) *Failure in Information Extraction*: Another cause of hallucinations is the failure to accurately extract relevant attributes or details in the model's higher-layer attention mechanisms. Known as "answer extraction hallucinations," this issue underscores the importance of precise information retrieval and application in generating correct outputs [15].

4) *Contextual Misalignment*: Contextual hallucinations occur when the model generates outputs misaligned with the provided context. These hallucinations are related to the model's difficulty in correctly attending to relevant context versus relying on its internal generative tendencies [11].

5) *Semantic Entropy*: The concept of semantic entropy has recently been proposed as a way to detect hallucinations caused by knowledge gaps in LLMs. This probabilistic measure of semantic equivalence helps identify hallucinations where the model's output strays from expected factual content due to insufficient knowledge [11].

C. Factors Influencing Hallucinations

Several factors contribute to the occurrence and severity of hallucinations in LLMs:

1) *Training Data Quality and Diversity*: The quality and diversity of training data are critical to the model's ability to produce accurate, contextually appropriate outputs. Imbalances or biases in the training data can lead to hallucinations, particularly in scenarios that are underrepresented during training [16].

2) *Model Architecture and Size*: The architecture and size of the LLM can influence its tendency to hallucinate. Larger models generally have more comprehensive knowledge but may also exhibit a higher propensity for hallucination due to increased complexity [12].

3) *Task Complexity*: The complexity of the task posed to the LLM also affects the likelihood of hallucinations. More complex or ambiguous queries are more likely to provoke hallucinated responses [17].

4) *Dynamic Nature of Language Model Reasoning*: Recent studies suggest that the reasoning processes within LLMs are dynamic, contributing to the complexity of understanding and mitigating hallucinations. This dynamic nature adds another layer of difficulty in precisely characterizing the mechanisms behind these errors [18].

D. Implications and Future Directions

Understanding the types and causes of hallucinations in LLMs is essential for enhancing their reliability and applicability in real-world scenarios. Future research should prioritize the following areas:

- **Robust evaluation methods**: Developing more comprehensive benchmarks, such as the HalluQA benchmark for Chinese LLMs, to systematically assess hallucination tendencies across different models and languages [17].
- **Mitigation strategies**: Exploring innovative techniques to reduce hallucinations, such as restoring the internal fact recall pipeline in language models to improve factual accuracy [15].
- **Long-chain reasoning approaches**: Investigating methods like long-chain reasoning to reduce hallucinations in high-stakes domains such as healthcare, legal advice, and education, where factual consistency is critical [19].
- **Dynamic reasoning studies**: Further examining the dynamic nature of language model reasoning to better understand the internal processes that contribute to hallucinations [18].

Addressing these challenges will be key to developing more trustworthy and accurate LLMs, allowing for safer and more effective deployment across a wide range of applications.

III. IMPACT OF HALLUCINATIONS

Hallucinations in Large Language Models (LLMs) are a topic of ongoing research and debate, presenting both risks and potential advantages across diverse domains. This section explores the consequences of hallucinations in high-stakes applications and highlights their potential benefits in creative processes, integrating recent findings and insights.

A. Consequences in Critical Applications

Hallucinations in LLMs pose substantial risks in domains where accuracy and reliability are essential, such as healthcare, scientific research, and information dissemination. In these areas, incorrect or fabricated outputs can have severe consequences, especially when integrated into decision-making processes.

In healthcare and scientific research, LLM hallucinations could lead to inaccurate diagnoses, inappropriate treatment suggestions, or misinterpretation of experimental data. Such errors not only endanger patient safety and research outcomes but also erode confidence in AI-assisted systems [20]. Perković et al. (2024) emphasize the need to thoroughly understand and address these issues to ensure the safe and effective deployment of LLMs in these critical fields [21].

The societal ramifications of hallucinations extend beyond individual applications. Hao et al. (2024) developed a framework to quantify the influence of AI hallucinations on social networks, illustrating the potential for LLM-generated misinformation to proliferate rapidly [22]. This underscores the urgent need for effective mitigation strategies to prevent the widespread dissemination of false information in complex social systems.

In addition, hallucinations may undermine LLM applications in fields requiring precision, such as motor skill learning. Qiu (2024) showed that hallucinations in LLM-guided skill acquisition could negatively affect performance in activities like sports, exemplified by the detrimental impact on badminton training [23]. These findings highlight the importance of carefully evaluating the suitability of LLMs in domains that demand physical precision and accuracy.

B. Potential Benefits in Creative Processes

Although typically viewed as a limitation, hallucinations in LLMs may also serve as a catalyst for creativity and innovation. Jiang et al. (2024) propose a novel perspective, suggesting that LLM hallucinations could facilitate creative thinking by generating unconventional ideas and associations [24].

In creative disciplines such as art, literature, and design, hallucinations can be likened to divergent thinking, where the model produces unique, unexpected concepts. This process can inspire new storylines, designs, or artistic expressions that might not emerge through conventional thought. For example, Yotam et al. (2024) introduced MyStoryKnight, a storytelling system that leverages LLM hallucinations to enhance creative narrative generation by integrating AI-driven character development with human-guided storytelling [25]. This showcases how LLMs can complement human creativity, resulting in novel and engaging experiences.

Furthermore, the recombination of information in novel ways, even if factually imprecise, can stimulate innovative problem-solving. This attribute is especially valuable in domains such as product design, where unconventional feature combinations may lead to groundbreaking innovations.

C. Balancing Risks and Benefits

The dual role of LLM hallucinations—as both a potential source of risk and creativity—requires a balanced and nuanced approach to their management and utilization. Oelschlager (2024) underscores the significant impact hallucinations can have on user trust and satisfaction in LLM-driven systems, emphasizing the importance of transparency and user education to mitigate these concerns [26]. As AI continues to be integrated into critical and everyday applications, preserving user confidence is essential to the technology’s broader acceptance.

In response to these challenges, researchers are developing various strategies to mitigate hallucinations. Tonmoy et al. (2024) present a comprehensive review of current hallucination mitigation techniques, offering insights into approaches that aim to reduce both the frequency and severity of hallucinations without compromising the generative strengths of LLMs

[20]. These strategies range from enhancing model training processes to developing more robust post-processing methods for validating outputs, ensuring a balance between minimizing risks and harnessing the creative potential of these models.

IV. DETECTION AND MITIGATION STRATEGIES

As Large Language Models (LLMs) become increasingly embedded in various applications, the issue of hallucinations—where models generate factually incorrect or contextually irrelevant information—has become a significant concern. This section discusses current approaches to detecting and mitigating hallucinations, focusing on detection methods, mitigation strategies, and grounding techniques.

A. Detection Methods

Detecting hallucinations in LLM outputs is a critical step in mitigating their impact. Several approaches have been developed to identify when an LLM produces unreliable or incorrect information:

- **Named Entity Recognition (NER) and Natural Language Inference (NLI):** Wang et al. propose a system that leverages NER and NLI techniques to detect a wide range of hallucinations in LLM outputs. This method identifies when the model generates content that contradicts the input or established facts [27].
- **Span-based Detection (SBD):** This method focuses on identifying specific spans of text within the LLM’s output that may contain hallucinated information. By analyzing smaller segments of generated content, SBD provides more granular insights into the occurrence of hallucinations [27].
- **Probability-based Methods:** Some researchers have developed frameworks that quantify the likelihood of hallucination in LLM outputs, generating confidence scores based on input and response attributes [28].
- **Two-Stage Detection Framework:** A novel approach uses smaller language models for initial hallucination detection, followed by larger LLMs that act as constrained reasoners to provide detailed explanations for detected hallucinations. This method aims to balance detection latency with interpretability and accuracy [29].
- **Hallucination Critic:** Luo et al. introduce a quantitative method that measures the percentage of hallucinated content in LLM outputs, offering a way to assess the extent of hallucination in generated text [30].
- **Entity-Level Hallucination Detection:** Su et al. propose a method designed to detect hallucinations at the entity level, specifically focusing on inconsistencies related to entities mentioned in the output [31].

B. Mitigation Strategies

After hallucinations are detected, various strategies can be employed to minimize their occurrence and impact:

- **Rewriting Mechanisms:** Rewriting mechanisms are designed to modify LLM outputs by correcting or removing hallucinated content, balancing precision, response time, and cost-effectiveness [27].

- **Multi-Scoring Framework:** Combining multiple scoring methods can enhance hallucination detection and mitigation. This approach leverages the strengths of various techniques, as no single scoring method performs optimally in all scenarios [28].
- **Prompt Engineering:** Carefully structuring input prompts can help reduce the likelihood of hallucinations. This may involve providing detailed context, specifying the format of the desired output, or explicitly instructing the model to avoid generating speculative information [20].
- **Fine-Tuning and Model Adjustments:** Fine-tuning LLMs on domain-specific datasets or adjusting model parameters can reduce hallucinations, especially in specialized applications [32].
- **Dynamic Retrieval Augmentation (DRAD):** Su et al. propose DRAD, a method that dynamically adjusts the retrieval process based on real-time hallucination detection to improve the factual accuracy of LLM responses [31].
- **Adversarial Testing:** Perković et al. recommend adversarial testing to identify LLM vulnerabilities and develop targeted strategies for mitigating specific types of hallucinations [21].

C. Grounding Techniques

Grounding techniques aim to anchor LLM outputs to reliable information sources, reducing the risk of hallucinations:

- **Retrieval-Augmented Generation (RAG):** RAG integrates information from trusted knowledge bases into the LLM's generation process. This ensures that the model's outputs are based on factual information rather than potentially hallucinated content [20].
- **Fact-Checking Integration:** Automated fact-checking systems that verify LLM outputs against reliable sources can help identify and correct hallucinations before the content is delivered to users [33].
- **Knowledge Graph Integration:** Incorporating structured knowledge graphs into the LLM's reasoning process provides a factual foundation for the model's responses, reducing hallucinations [20].
- **External API Calls:** For domain-specific applications, integrating LLMs with external APIs that provide real-time, accurate information helps ground outputs in trustworthy data [32].
- **Self-Contradiction Detection:** Recent research focuses on detecting and addressing self-contradictions within LLM outputs, a common form of hallucination where the model generates internally inconsistent information [20].
- **Vision-Language Model Hallucination Mitigation:** As LLMs expand into multimodal applications, researchers are developing techniques to address hallucinations in vision-language models, particularly aligning visual content with textual generation to prevent inconsistencies [12].

By combining these detection methods, mitigation strategies, and grounding techniques, researchers aim to enhance the reliability and trustworthiness of LLM-generated content across

a wide range of applications. As the field progresses, new methods and refinements to existing strategies will likely emerge, further improving the ability to manage hallucinations in LLMs.

V. FUTURE DIRECTIONS AND CONCLUSION

As Large Language Models (LLMs) continue to advance and become embedded in various applications, addressing the issue of hallucinations remains a key research focus. This section summarizes the current state of research on LLM hallucinations and outlines future directions to improve their reliability and ethical application.

A. Improved Evaluation Methods

Developing rigorous evaluation criteria that align with expert perspectives is crucial for detecting and mitigating hallucinations in LLMs. Current research underscores the need for comprehensive evaluation frameworks capable of accurately identifying hallucinations across different tasks and domains.

Chakraborty et al. (2024) emphasize the need for flexible definitions and context-aware evaluation methods for hallucination detection, especially in decision-making settings [34]. They suggest future research should focus on refining evaluation techniques that can distinguish between types of hallucinations and assess their potential impact on decision-making processes.

Schiller (2024) advocates for incorporating the human factor into error detection, proposing that understanding how users identify and interpret hallucinations could lead to more effective evaluation strategies [35]. This human-centered approach may help bridge the gap between technical evaluations and real-world applications.

B. Ethics and Responsible AI

Balancing the creative potential of hallucinations with ethical standards remains a critical challenge for future research. The ethical implications of hallucinations go beyond accuracy concerns, touching on trust, accountability, and societal impacts.

Jiao et al. (2024) call for ethical frameworks tailored to specific domains and dynamic auditing systems that adapt to diverse contexts [36]. They emphasize the need for interdisciplinary collaboration to create guidelines that can keep pace with the rapid evolution of LLM technology.

Lin et al. (2024) highlight the importance of developing debiasing and dehallucination techniques to build more trustworthy LLMs [37]. Future work should focus not only on reducing hallucinations but also on ensuring that LLM outputs align with ethical standards and societal values.

C. Advanced Integration Techniques

Balancing external knowledge with the internal knowledge encoded in LLMs is essential for reducing hallucinations while preserving generative capabilities. Future research should explore more sophisticated integration methods that leverage the strengths of both knowledge sources.

Tonmoy et al. (2024) review hallucination mitigation strategies and propose that future research should explore advanced

techniques for integrating external knowledge with LLMs' parametric memory [20]. Techniques such as dynamic knowledge retrieval and adaptive fusion mechanisms hold promise for improving the reliability of LLM outputs.

Bai et al. (2024) focus on hallucinations in multimodal LLMs, calling for further exploration of integration techniques that effectively combine textual and visual data to minimize hallucinations in multimodal tasks [13]. Research in this area could pave the way for more robust and reliable multimodal AI systems.

D. Conclusion

In summary, research on hallucinations in Large Language Models (LLMs) has made significant strides in understanding, detection, and mitigation. Future efforts should prioritize refining evaluation methods, addressing ethical considerations, and advancing integration techniques. By focusing on these areas, the goal is to create more reliable, trustworthy, and ethically responsible LLMs suitable for a wide range of applications.

As LLM technology evolves, collaboration among researchers, developers, and policymakers will be critical to ensure its responsible development and deployment. Emphasizing the minimization of hallucinations and maximizing the benefits of LLMs will be key to unlocking their full potential while managing the associated risks.

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